

5 Two small coherent sound sources S_1 and S_2 are set up as shown in Fig.5.1 below.

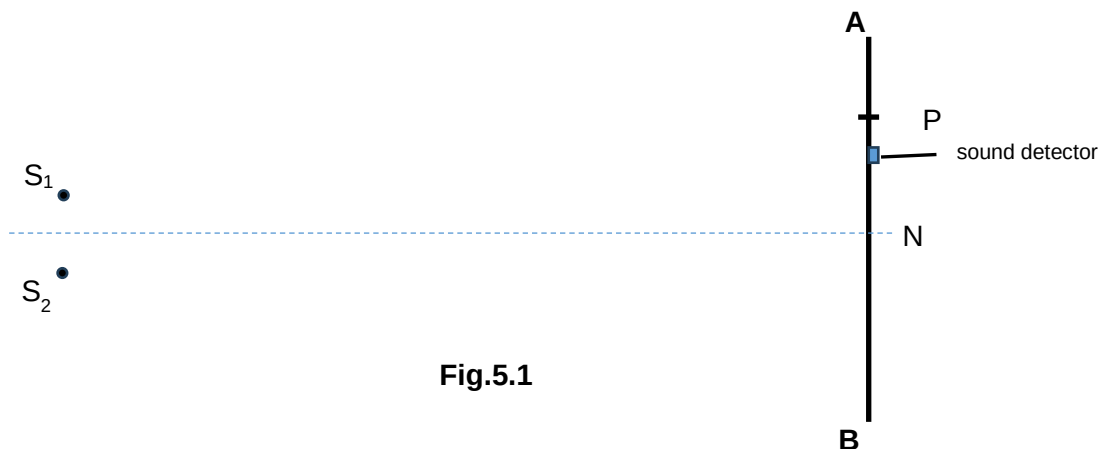


Fig.5.1

A sound detector is moved along a line AB that is parallel to S_1S_2 . N is the point on AB such that $S_1N = S_2N$. The sound waves from S_1 and S_2 have frequency 2.80 kHz and speed 336 m s⁻¹.

(a) Show that the wavelength of the waves is 12.0 cm.

[1]

Using the wave equation $v=f\lambda$,

$$\lambda = v/f = 336/2.8 \times 10^3$$

$$= 0.120 \text{ m} = 12.0 \text{ cm (shown)}$$

(b) The detector, when placed at N, indicates a maximum intensity of sound. As it is moved from N to a point P, the intensity varies between high and low values. At P, the distance S_1P is 372 cm and S_2P is 402 cm.

Determine, with suitable explanation,

(i)	whether the intensity of sound at P is high or low,
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[2]

Path difference = $402 - 372 = 30 \text{ cm} = (2 \frac{1}{2}) \lambda$. [B1]

Given that a maximum intensity occurs at N, we can infer that sources S_1 and S_2 are producing waves in phase with each other.

Hence a path difference of odd integer multiple of half-wavelength implies that the waves arrive at P in antiphase. They would superpose destructively producing sound of low intensity. [B1]

(ii)	the number of high intensity regions that are found between N and P. Do not include the maximum at N.
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			 [3]
			From N to P, path difference increases from zero to $(2 \frac{1}{2}) \lambda$. [B1] This means that there are two high intensity regions where constructive interference occurs. [B1] They are located at the region where the path difference is 1λ and 2λ. [B1]
			[Total: 6]

6	The circuit is used to supply energy to the battery from the generator. A variable resistor of
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